

*MA.7.DP.2.3***Benchmark**

MA.7.DP.2.3 Find the theoretical probability of an event related to a simple experiment.

Benchmark Clarifications:

Clarification 1: Instruction includes representing probability as a fraction, percentage or decimal.

Clarification 2: Simple experiments include tossing a fair coin, rolling a fair die, picking a card randomly from a deck, picking marbles randomly from a bag and spinning a fair spinner.

Connecting Benchmarks/Horizontal Alignment

- MA.7.AR.3.1
- MA.7.DP.2.1

Terms from the K-12 Glossary

- Event
- Theoretical Probability

Vertical Alignment**Previous Benchmarks**

- MA.6.NSO.3.5

Next Benchmarks

- MA.8.DP.2.3

Purpose and Instructional Strategies

In grade 7, students are introduced to probability and will find the theoretical probability of an event related to a simple experiment. In grade 8, students will find the theoretical probability of an event related to a repeated experiment.

- Instruction builds on finding sample spaces from MA.7.DP.2.1. Have students discuss their understanding of the words “theoretical” and “probability” to build toward a formal definition of theoretical probability.
- When finding theoretical probability, have students work from their sample space. Doing so will lead to the understanding that since experiments for this benchmark are fair, the probability of an event is equivalent to $\frac{\text{number of outcomes in the event}}{\text{number of outcomes in the sample space}}$.
 - For example, if rolling a fair 6-sided die, the sample space is $\{1, 2, 3, 4, 5, 6\}$. If one wants to find $P(\text{rolling an odd number})$, students can circle all of the odd numbers from the sample space to determine the probability as $\frac{3}{6}$, or 0.5.
- While the benchmark does focus on fair experiments, instruction could include spinners with unequal sections making the connection to angle measures and to circle graphs.
- Instruction focuses on the simple experiments listed in *Clarification 2*.
 - For example, when tossing a coin with one yellow side and one red side, $P(\text{landing on blue}) = 0$.
 - For example, when rolling a 10-sided die, $P(\text{not rolling a multiple of 3}) = 0.7$.
 - For example, when picking a card from a deck that contains each of the letters of the alphabet, $P(\text{picking a consonant}) = \frac{21}{26}$.
 - For example, when picking a tile from a bag that contains a set of chess pieces,

$P(\text{picking a pawn}) = 50\%$.

- For example, when spinning a spinner that contains 5 sections where two of the sections are green and the remaining sections are red, white and blue,

$P(\text{landing on a color from the American flag}) = \frac{3}{5}$.

Common Misconceptions or Errors

- Students may incorrectly assume that an event is only possible in an experiment and interchange the meaning of an event and an experiment.
- Students may confuse the mathematical meaning of a word like “event” with the everyday meaning.
- Students may incorrectly convert forms of probability between fractions and percentages. To address this misconception, scaffold with more familiar values initially to facilitate the interpretation.
- Students may incorrectly interpret a value with a negative sign as a possible probability.
 - For example, $-\frac{1}{2}$ cannot represent a probability since negative values are less than 0.

Strategies to Support Tiered Instruction

- Teacher provides instruction in converting between fractions and percentages, by using more familiar values.
 - For example, $\frac{1}{2} = 50\%$, $\frac{1}{4} = 25\%$, $\frac{1}{3} \approx 33.3\%$, etc.
- Teacher co-creates a T-chart (like the one below) to list the experiment and the sample space necessary for the examples provided.

Experiment	Sample Space
Tossing a coin	H, T
Drawing a marble from a bag containing 3 red marbles and 2 green marbles	R1, R2, R3, G1, G2 or R, R, R, G, G
Rolling a 6-sided die	1, 2, 3, 4, 5, 6

Instructional Tasks

Instructional Task 1 (MTR.4.1, MTR.7.1)

Look at the shirt you are wearing today and determine how many buttons it has. Then complete the following table using tally marks for all the members of your class.

	No Buttons	One or Two Buttons	Three or Four Buttons	More Than Four Buttons
Male				
Female				

Suppose each student writes his or her name on an index card, and one card is selected randomly.

Part A. What is the probability that the student whose card is selected is wearing a shirt with no buttons?

Part B. What is the probability that the student whose card is selected is female and is wearing a shirt with two or fewer buttons?

Instructional Task 2 (MTR.4.1, MTR.5.1)

There is only one question on the next quiz, and it will be true or false.

Part A. If a student randomly answers the question, what is the probability of earning a score of 100%?

Part B. What is the probability of earning a 50%?

Instructional Items

Instructional Item 1

What is the probability of choosing a 9 from a standard deck of 52 cards?

Instructional Item 2

There are 7 red, 5 blue and 12 green marbles in a bag.

Part A. What is the probability of choosing a red marble?

Part B. What is the probability of not choosing a green marble?

Part C. What is the probability of choosing a yellow marble?

Part D. What is the probability of choosing a red or blue marble?

**The strategies, tasks and items included in the BIG-M are examples and should not be considered comprehensive.*